

Sara Zhiliu He . Jarvis Consulting

I earned a Master's degree in Systems Design Engineering (Artificial Intelligence and Machine Learning Specialization) from the University of Waterloo, where I built great foundation for software engineering and AI. Eager to expand my skill set, I completed a software engineering diploma program at BrainStation, gaining hands-on experience in full-stack web development. I am an energetic software developer with professional experience as a Software Engineer at Jarvis and a Full-Stack Developer at Skill4Skill, driven by a passion for building intuitive and scalable web applications that deliver real value to end users and solve real-world challenges.

Skills

Proficient: Java, Python, Linux/Bash, RDBMS/SQL, Docker, JavaScript/TypeScript, React/Next.js, Node.js/Express, Agile/Scrum, Git

Competent: Spring Boot, Angular, MongoDB, CI/CD, Pandas/NumPy

Familiar: AI/Machine Learning, TensorFlow, PyTorch, Scikit-Learn, Hadoop, Spark, Figma

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_SarazhiliuHe

Linux Cluster Monitoring Agent [GitHub]: Built the Jarvis Linux Cluster Monitoring Agent to assist the Linux Cluster Administration (LCA) team in managing a multi-node Rocky Linux environment. Implemented a Bash Script to collect static system information, such as CPU architecture, memory, and disk specs, executed once on each node. Automated the collection of real-time resource usage data, such as CPU, memory, and disk usage, using a Bash script triggered by a Cron job every minute. Deployed a PostgreSQL database in a Docker container to store the collected data in a portable environment. Used Git and GitHub for source code management and version control.

Java Grep App [GitHub]: Developed Grep App, a Java application that mimics the Linux grep command to search for lines matching a regular expression across a directory of text files. Leveraged Java 8 Lambda and Stream APIs for efficient and readable data processing. Utilized Maven for project build and dependency management. Dockerized the application for seamless distribution and deployment.

Python Data Analytics [GitHub]: Developed a Proof of Concept (PoC) data analysis project for LGS, a retail company, to help the marketing team optimize campaign strategies. Designed and provisioned a local PostgreSQL OLAP data warehouse using Docker to ingest and organize raw retail data. Performed in-depth sales performance and customer behavior analysis using Python (Pandas, NumPy) in Jupyter Notebook, identifying key purchasing patterns and trends. Delivered insights and visualizations via GitHub to support data-driven decision-making.

Spark [GitHub]: Developed a Hadoop-based analytics workflow using Google Cloud Dataproc and Apache Zeppelin. Exported the WDI dataset from BigQuery to Google Cloud Storage and transformed raw CSV files into optimized Parquet tables in HDFS using HiveQL. Performed exploratory and complex analytics using PySpark within Zeppelin notebooks. Provisioned an Azure Databricks workspace with a Spark cluster to analyze LGS retail data using PySpark Structured APIs.

Highlighted Projects

MasterMind [GitHub]: Developed and maintained a web application for educational content creators using Next.js and TypeScript, and created reusable UI components to enhance consistency and maintainability. Designed and optimized PostgreSQL databases using Prisma ORM for seamless and efficient data interactions. Implemented security best practices, including role-based access control, and integrated secure authentication with Clerk. Deployed the application on Vercel, ensuring fast load times and global availability.

InStock: Built a responsive warehouse inventory management website using HTML, CSS, React, and JavaScript, adhering to functional and visual design requirements. Developed and managed APIs with Node.js, Express.js, and Knex.js, optimizing database interactions with MySQL. Handled version control and team collaboration using Git and GitHub, utilized Jira for project management, and participated in code reviews to ensure high-quality development and on-time delivery.

Professional Experiences

Software Developer, Jarvis (APR 2025-present): Collaborated within an Agile team to deliver diverse software development projects across multiple sprints. Developed backend tools and automation scripts using Linux/Bash, SQL, Java, and Python to support data processing and system maintenance tasks. Worked with Google Cloud Platform (GCP) to provision virtual machines and access company infrastructure via remote desktop. Used Git for version control and Docker for containerization to build, test, and maintain scalable applications.

Full Stack Software Developer, Skill4Skill (MAR 2025-present): Collaborated with cross-functional teams to develop Skill4Skill, a web platform where individuals can exchange expertise. Built a responsive and accessible frontend in Next.js, ensuring alignment with design specifications and style guides, and worked closely with designers to refine UI/UX. Designed MongoDB schemas and developed RESTful APIs with Express.js, and integrated backend APIs with frontend components for seamless feature delivery. Maintained documentation for codebases, workflows, and API specifications. Participated in Agile development using Jira for sprint planning, user story tracking, and backlog management.

Research Assistant, University of Waterloo (MAY 2024-DEC 2024): Researched and developed mathematical optimization models to improve multi-echelon supply chain performance, focusing on minimizing costs and improving inventory allocation efficiency. Applied Python, Pandas, NumPy, and Gurobi to build and test mixed-integer linear programming (MILP) models simulating real-world supply chain scenarios. Drafted detailed analytical reports summarizing methodology, results, and analyses. Collaborated closely with the supervising professor to iteratively refine the research paper.

Education

University of Waterloo (2023-2024), Master of Engineering, Systems Design Engineering - GPA: 85/100 - Specialization in Artificial Intelligence and Machine Learning

South China University of Technology (2019-2023), Bachelor of Engineering, Logistics Engineering - GPA: 92/100 - First-class Academic Scholarship - Student Science and Technology Innovation Competition Scholarship - PG Logistics Scholarship - First prize in the National Contest in Big Data Analysis and Mining - Third Prize in the China Undergraduate Mathematical Contest in Modeling

Miscellaneous

- BrainStation Software Engineering Diploma (2024)
- Applied Python Data Engineering Specialization, Duke University (Coursera), May 2024
- Google Data Analytics Specialization, Google (Coursera), May 2024
- Dancing: Captain of Dance Team at South China University of Technology
- Volunteer Work: Member of Campus Volunteer Service Team (stray cat rescue and campus waste recycling)
- Reading: Avid reader and fan of Agatha Christie's detective fiction.